

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Machine Learning

Exam: IN2064 / Endterm

Date: Thursday 12th February, 2026

Examiner: Prof. Dr. Stephan Günnemann

Time: 08:00 – 10:00

	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8	P 9
I									

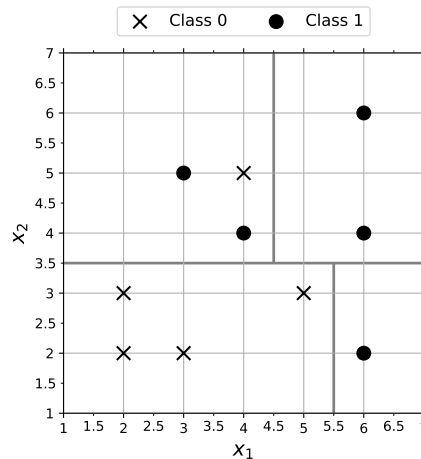
Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 44 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam.
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____

Problem 1 Decision Trees and kNN (4 credits)

Consider the following decision boundary of a decision tree trained on a two-dimensional dataset with features $x_1, x_2 \in \mathbb{R}$ and two classes:



- 0 ☐
1 ☐
- a) Compute the improvement $\Delta(s, t)$ with respect to the first split s at root node t using the misclassification error $i_E(t)$ as impurity measure. Justify your answer.

$$\Delta(s, t) = i_E(t) - p_L i_E(t_L) - p_R i_E(t_R)$$

$$i_E(t) = 1 - \max_c \pi_c = \frac{1}{2}$$

$$i_E(t_L) = \frac{1}{5} \quad i_E(t_R) = \frac{1}{5}$$

$$p_L = \frac{1}{2} \quad p_R = \frac{1}{2}$$

$$\Rightarrow \Delta(s, t) = \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{5} - \frac{1}{2} \cdot \frac{1}{5} = \frac{1}{2} - \frac{1}{5} = \frac{3}{10}$$

- 0 ☐
1 ☐
- b) Name two problems of using the misclassification error as impurity measure in decision tree learning.

1. May lead to too early stopping of the tree building due to zero impurity improvement, even though splitting would lead to better class separation.
2. Not sensitive to changes in class probabilities in the nodes.

- 0 ☐
1 ☐
2 ☐
- c) Consider an online learning scenario in which new test points are continuously classified. Additionally, consider now that a small number of training samples are updated over time (e.g. their position or label change). Explain how the decision trees discussed in the lecture and k-nearest neighbor classifiers respond to such changes, and discuss which classifier is more suitable in this setting and why. Justify your answer.

Decision trees are trained by recursively splitting the training data. Even small changes in the training set can alter optimal splits (optimal under a splitting heuristic), which typically requires rebuilding the entire tree to correct the decision boundary.

kNN does not have a training phase, no model needs to be rebuilt. Only requires recomputation of the distances to the changed training points.

Thus kNN is preferable, as incremental changes can be handled more efficiently without retraining.

Problem 2 Probabilistic Inference (6 credits)

Let $X \in \{0, 1\}$ be a random variable denoting the result of a (possibly biased) coin flip ($X = 0$ for tails and $X = 1$ for heads). Let θ_1 denote the probability of observing heads. Suppose that you have a smart device that looks at the flipped coin and says its outcome, which we denote as Y . However, this device is not always reliable and only says the correct outcome with probability θ_2 . Formally, we have

$$p(y|x, \theta_2) = \begin{cases} \theta_2, & \text{if } y = x \\ 1 - \theta_2, & \text{otherwise.} \end{cases}$$

We assume that θ_2 is independent of θ_1 .

a) Write down the joint probability distribution $p(x, y|\theta)$ in terms of the parameter vector $\theta = (\theta_1, \theta_2)^T$ for all possible values of x and y .

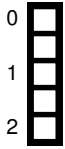
0
1

The joint distribution can be factorized as follows:

$$p(x, y|\theta) = p(x|\theta_1)p(y|x, \theta_2).$$

We can visualize it as a 2x2 table for all possible values of x and y .

	$y = 0$	$y = 1$
$x = 0$	$(1 - \theta_1)\theta_2$	$(1 - \theta_1)(1 - \theta_2)$
$x = 1$	$\theta_1(1 - \theta_2)$	$\theta_1\theta_2$



b) Suppose you collect the following dataset of (x, y) pairs:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n = \{(1, 1), (1, 0), (0, 0), (0, 1), (1, 1), (0, 0), (0, 1)\}.$$

Derive the maximum likelihood estimates (MLEs) $\hat{\theta}_1$ and $\hat{\theta}_2$ for the 2 parameters θ_1 and θ_2 .

We can represent the dataset as a count table:

	$y = 0$	$y = 1$
$x = 0$	2	2
$x = 1$	1	2

With that, we can write down the likelihood function:

$$\begin{aligned} p(\mathcal{D}|\theta) &= \prod_{i=1}^n p(x_i, y_i|\theta) \\ &= (1 - \theta_1)^2 \theta_2^2 (1 - \theta_1)^2 (1 - \theta_2)^2 \theta_1 (1 - \theta_2) \theta_1^2 \theta_2^2 \\ &= \theta_1^3 (1 - \theta_1)^4 \theta_2^4 (1 - \theta_2)^3 \end{aligned}$$

$$\log p(\mathcal{D}|\theta) = 3 \log \theta_1 + 4 \log(1 - \theta_1) + 4 \log \theta_2 + 3 \log(1 - \theta_2)$$

To get the minimizer, we compute the derivative and set it to 0:

$$\begin{aligned} \frac{\partial \log p(\mathcal{D}|\theta)}{\partial \theta_1} &= \frac{3}{\theta_1} - \frac{4}{1 - \theta_1} \stackrel{!}{=} 0 \\ \Leftrightarrow 4\theta_1 &= 3 - 3\theta_1 \\ \Leftrightarrow \hat{\theta}_1 &= \frac{3}{7} \end{aligned}$$

Similarly, we get $\hat{\theta}_2 = \frac{4}{7}$.



c) Now instead of the previous 2-parameter model, consider a model with 4 parameters $\theta = (\theta_{0,0}, \theta_{0,1}, \theta_{1,0}, \theta_{1,1})$ representing $p(x, y|\theta) = \theta_{x,y}$, where $\theta_{0,0} + \theta_{0,1} + \theta_{1,0} + \theta_{1,1} = 1$. Write down the likelihood function of this model on the observed dataset $\mathcal{D}$.

$$\begin{aligned} p(\mathcal{D}|\theta) &= \prod_{i=1}^n p(x_i, y_i|\theta) \\ &= \theta_{0,0}^2 \theta_{0,1}^2 \theta_{1,0} \theta_{1,1}^2 \end{aligned}$$



d) Suppose you are not sure which of the two models to use. So, you decide to compute the leave-one-out cross validated log likelihood of both models as follows: $L(m) = \sum_{i=1}^n \log p(x_i, y_i | m, \hat{\theta}(\mathcal{D}_{-i}))$, where m selects one of the two models and $\hat{\theta}(\mathcal{D}_{-i})$ denotes the MLE of the model parameters computed on $\mathcal{D}$ excluding the i -th pair. Which model is preferred according to this metric? **Justify** your answer.

Hint 1: For the 4-parameter model, the MLEs are given as $\theta_{x,y} = \frac{N_{(x,y)}}{N}$, where $N_{(x,y)}$ is the number of pairs in $\mathcal{D}$ having the values (x, y) and N is the total size of $\mathcal{D}$.

Hint 2: Specifically consider the term where the second training sample is left out.

For the 4-parameter model, when we omit the second sample $(1, 0)$, we will have $\hat{\theta}_{1,0} = 0$, so $p(x_1, y_1 | m_4, \hat{\theta}(\mathcal{D}_{-2})) = 0$ and thus $L(m_4) = -\infty$. On the other hand, $L(m_2)$ will be finite in all cases. Hence, the 2-parameter model is preferred since the 4-parameter model is overfitting.

Problem 3 Poisson Regression (4 credits)

In this task, we consider a regression problem where the target variable is a count variable, e.g., the number of emails expected on a specific day. In this case, a common choice for the likelihood function is the Poisson distribution, which has its support over positive integers. Specifically, a random variable $Y \in \mathbb{N}_0$ has a Poisson distribution with rate $\lambda > 0$ if its probability mass function is given by:

$$p(y|\lambda) = \frac{e^{-\lambda} \lambda^y}{y!}.$$

In Poisson regression, instead of assuming a fixed λ , we model it as a function of some input data $\mathbf{x} \in \mathbb{R}^d$, e.g., the day of the week might influence the number of expected emails on that day. Specifically, we assume $\log \lambda = \boldsymbol{\theta}^\top \mathbf{x}$, where $\boldsymbol{\theta} \in \mathbb{R}^d$ is the parameter vector.

- 0 ☐ a) Consider a training dataset consisting of features $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$ with respective count targets $\mathbf{y} = (y_1, \dots, y_n)^\top$, where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \mathbb{N}_0$. Write down the negative log-likelihood, $-\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta})$, of a Poisson regression model with parameter $\boldsymbol{\theta} \in \mathbb{R}^d$ in terms of $\boldsymbol{\theta}$, $\mathbf{x}_i$ and y_i .
- 1 ☐

$$\begin{aligned} -\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) &= -\log \prod_{i=1}^n p(y_i|\mathbf{x}_i, \boldsymbol{\theta}) \quad (\text{i.i.d}) \\ &= -\sum_{i=1}^n \log \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}, \quad \lambda_i = e^{\boldsymbol{\theta}^\top \mathbf{x}_i} \\ &= -\sum_{i=1}^n -\lambda_i + y_i \log \lambda_i - \log(y_i!) \\ &= -\sum_{i=1}^n -e^{\boldsymbol{\theta}^\top \mathbf{x}_i} + y_i \boldsymbol{\theta}^\top \mathbf{x}_i - \log(y_i!) \\ &= \sum_{i=1}^n e^{\boldsymbol{\theta}^\top \mathbf{x}_i} - y_i \boldsymbol{\theta}^\top \mathbf{x}_i + \log(y_i!) \end{aligned}$$

☐	0
☐	1
☐	2

b) To find the optimal parameters θ^* , we need to minimize the negative log-likelihood. However, this optimization problem has no closed-form solution, so we need to use gradient-based methods instead. **Derive** the gradient descent update rule for the j -th parameter θ_j of a Poisson regression model. Assume a learning rate $\alpha \in \mathbb{R}$.

$$\begin{aligned}
 \frac{\partial(-\log p(\mathbf{y}|\mathbf{X}, \theta))}{\partial \theta_j} &= \frac{\partial(\sum_{i=1}^n e^{\theta^\top \mathbf{x}_i} - y_i \theta^\top \mathbf{x}_i + \log(y_i!))}{\partial \theta_j} \\
 &= \sum_{i=1}^n \frac{\partial(e^{\theta^\top \mathbf{x}_i})}{\partial \theta_j} - \frac{\partial(y_i \theta^\top \mathbf{x}_i)}{\partial \theta_j} + \frac{\partial(\log(y_i!))}{\partial \theta_j} \\
 &= \sum_{i=1}^n e^{\theta^\top \mathbf{x}_i} x_{i,j} - y_i x_{i,j} \\
 &= \sum_{i=1}^n (e^{\theta^\top \mathbf{x}_i} - y_i) x_{i,j}
 \end{aligned}$$

Gradient update rule:

$$\begin{aligned}
 \theta_j &:= \theta_j - \alpha \frac{\partial(-\log p(\mathbf{y}|\mathbf{X}, \theta))}{\partial \theta_j} \\
 &= \theta_j - \alpha \sum_{i=1}^n (e^{\theta^\top \mathbf{x}_i} - y_i) x_{i,j}.
 \end{aligned}$$

☐	0
☐	1

c) What are the advantages of using a Poisson regression model for a count prediction problem over (i) a linear regression model, and (ii) a linear classification model? Name one advantage for each case.

- (i) A Poisson regression model ensures that the predicted values are non-negative, as the rate parameter λ is modeled as an exponential function of the input features. In contrast, a linear regression model can produce negative predictions, which are not meaningful for count data.
- (ii) A Poisson regression model produces count values up to infinity, while a linear classification model typically outputs discrete class labels. Using a classification model for count data would require discretizing the counts into classes, which can lead to loss of information and reduced predictive performance.

Problem 4 Linear Classification (4 credits)

Consider a binary classification problem on the feature space $x \in \mathbb{N}_0$ of natural numbers (including 0) with labels $y \in \{0, 1\}$. You are given a probabilistic generative classifier that models the prior class probabilities as

$$p(y = 0) = \frac{4}{5} \quad \text{and} \quad p(y = 1) = \frac{1}{5},$$

and the class-conditional feature distribution as a Poisson distributions, i.e.,

$$p(x | y = k) = \frac{e^{-\lambda_k} \lambda_k^x}{x!}$$

for $k \in \{0, 1\}$, $\lambda_0 \neq \lambda_1$, and $\lambda_i > 0$.

a) Classify the point $x = 2$ given $\lambda_0 = 1$ and $\lambda_1 = 2$. **Justify your answer!**

Using Bayes' theorem, we have

$$p(y = 0 | x) \propto p(x | y = 0)p(y = 0) = \frac{e^{-1} 1^2}{2!} \cdot \frac{4}{5} = \frac{2}{5} e^{-1}$$

and

$$p(y = 1 | x) \propto p(x | y = 1)p(y = 1) = \frac{e^{-2} 2^2}{2!} \cdot \frac{1}{5} = \frac{2}{5} e^{-2}.$$

Since $e^{-1} > e^{-2}$, we have $p(y = 0 | x) > p(y = 1 | x)$, and thus we classify the point $x = 2$ as belonging to class 0: $\arg \max_{k \in \{0, 1\}} p(y = k | x) = 0$.

b) Derive the decision boundary of this classifier in terms of λ_0 and λ_1 . **Justify your answer!**

We have to set $p(y = 0 | x) = p(y = 1 | x)$ to find the decision boundary. Using Bayes' theorem, and since the PMF of the Poisson distribution is non-zero, this is equivalent to setting

$$\ln \frac{p(x | y = 1)p(y = 1)}{p(x | y = 0)p(y = 0)} \stackrel{!}{=} 0.$$

$$\Leftrightarrow \ln \frac{p(x | y = 1)}{p(x | y = 0)} + \ln \frac{1}{4} = 0$$

$$\Leftrightarrow \ln \frac{e^{-\lambda_1} \lambda_1^x / x!}{e^{-\lambda_0} \lambda_0^x / x!} = \ln 4$$

$$\Leftrightarrow \lambda_0 - \lambda_1 + x \ln \frac{\lambda_1}{\lambda_0} = \ln 4$$

$$\Leftrightarrow x = \frac{\ln 4 + \lambda_1 - \lambda_0}{\ln \frac{\lambda_1}{\lambda_0}}.$$

Problem 5 Optimization (6 credits)

Recall that the negative log-likelihood (NLL) of a logistic regression model can be written as

$$E(\mathbf{w}) = -\log p(\mathbf{y}|\mathbf{X}, \mathbf{w}).$$

Definition: A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called **strongly convex** if

$$f(\mathbf{y}) - f(\mathbf{x}) \geq (\mathbf{y} - \mathbf{x})^T \nabla_{\mathbf{x}} f(\mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

for a parameter $\mu > 0$ and all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$.

Note that for $\mu = 0$, this definition reduces to the definition of convexity as in the lecture.

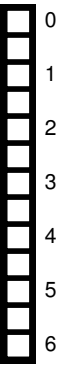
Interestingly, one can show that strongly convex functions have a unique global minimum. We want to use this to show that the NLL of a logistic regression model with L2-regularization has a unique global minimum.

Prove that the function $h(\mathbf{w}) = E(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$ is strongly convex for any $\lambda > 0$.

Hint: You may use without proof that the negative log-likelihood $E(\mathbf{w})$ is a convex function on $\mathbb{R}^d$.

$$\begin{aligned} & (\mathbf{y} - \mathbf{x})^T \nabla h(\mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \\ &= (\mathbf{y} - \mathbf{x})^T \nabla E(\mathbf{x}) + (\mathbf{y} - \mathbf{x})^T \nabla \left(\frac{\lambda}{2} \|\mathbf{x}\|_2^2 \right) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \\ &= (\mathbf{y} - \mathbf{x})^T \nabla E(\mathbf{x}) + \lambda (\mathbf{y} - \mathbf{x})^T \mathbf{x} + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \\ &\leq E(\mathbf{y}) - E(\mathbf{x}) + \lambda (\mathbf{y} - \mathbf{x})^T \mathbf{x} + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \quad (\text{since } E \text{ is convex}) \\ &= E(\mathbf{y}) - E(\mathbf{x}) + \lambda (\mathbf{y} - \mathbf{x})^T \mathbf{x} + \frac{\mu}{2} (\mathbf{y} - \mathbf{x})^T (\mathbf{y} - \mathbf{x}) \\ &= E(\mathbf{y}) - E(\mathbf{x}) + \lambda \mathbf{y}^T \mathbf{x} - \lambda \|\mathbf{x}\|_2^2 + \frac{\mu}{2} \|\mathbf{y}\|_2^2 - \mu \mathbf{y}^T \mathbf{x} + \frac{\mu}{2} \|\mathbf{x}\|_2^2 \\ &= E(\mathbf{y}) - E(\mathbf{x}) + \frac{\lambda}{2} \|\mathbf{y}\|_2^2 - \frac{\lambda}{2} \|\mathbf{x}\|_2^2 \quad (\text{choosing } \mu = \lambda) \\ &= h(\mathbf{y}) - h(\mathbf{x}) \end{aligned}$$

Thus, $h(\mathbf{w})$ is strongly convex with parameter $\mu = \lambda > 0$.



Problem 6 Deep Learning (6 credits)

Consider a fully connected neural network with one hidden layer of K neurons. The network as a whole is defined by this function:

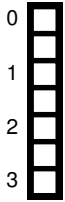
$$f(\mathbf{x}) = \sum_{k=1}^K v_k \text{ReLU}(\mathbf{W}_k^\top \mathbf{x} + b_k),$$

where $\mathbf{x} \in \mathbb{R}^D$ is the input, $v_k, b_k \in \mathbb{R}$, and $\mathbf{W} \in \mathbb{R}^{K \times D}$ is the weight matrix of the hidden layer.

$\mathbf{W}_k \in \mathbb{R}^{1 \times D}$ denotes the k -th row of $\mathbf{W}$.

The activation function is:

$$\text{ReLU}(t) = \max(0, t), \quad \text{ReLU}'(0) = 0.$$



a) **Derive** the partial derivative $\frac{\partial f(\mathbf{x})}{\partial x_d}$, for $d \in \{1, \dots, D\}$.

For $d \in \{1, \dots, D\}$, we compute the partial derivative

$$\frac{\partial f(\mathbf{x})}{\partial x_d} = \sum_{k=1}^K v_k \text{ReLU}'(\mathbf{W}_k^\top \mathbf{x} + b_k) \frac{\partial (\mathbf{W}_k^\top \mathbf{x} + b_k)}{\partial x_d}.$$

Since

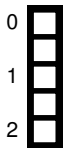
$$\frac{\partial (\mathbf{W}_k^\top \mathbf{x} + b_k)}{\partial x_d} = W_{k,d},$$

and

$$\text{ReLU}'(t) = \mathbf{1}[t > 0],$$

we obtain

$$\frac{\partial f(\mathbf{x})}{\partial x_d} = \sum_{k=1}^K v_k \mathbf{1}[\mathbf{W}_k^\top \mathbf{x} + b_k > 0] W_{k,d}.$$



b) How many distinct values can the gradient $\nabla_{\mathbf{x}} f(\mathbf{x}) \in \mathbb{R}^D$ take for $\mathbf{x} \in \mathbb{R}^D$? **Justify your answer!**

Every hidden unit is either active or inactive:

$$a_k(\mathbf{x}) = \mathbf{1}[\mathbf{W}_k^\top \mathbf{x} + b_k > 0] \in \{0, 1\}$$

Every $(a_1(\mathbf{x}), \dots, a_K(\mathbf{x}))$ corresponds to a single gradient value.
Therefore, the gradient can take at most 2^K distinct values.



c) How many distinct values can the gradient $\nabla_{\mathbf{x}} f(\mathbf{x}) \in \mathbb{R}^D$ take if we replace the ReLU activation function by $\text{LeakyReLU}(t) = \max(\alpha t, t)$, $\alpha > 0$?
We assume that $\text{LeakyReLU}'(0) = \alpha$.

$$\frac{d}{dt} \text{LeakyReLU}(t) \begin{cases} 1, & t > 0, \\ \alpha, & t \leq 0, \end{cases}$$

Every hidden unit takes values in $\{1, \alpha\}$. The gradient can take at most 2^K distinct values.

Problem 7 Support Vector Machines (7 credits)

Consider the soft-margin SVM primal problem:

$$\min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i.$$

a) Write down the dual optimization problem. What are the constraints on the Lagrange multipliers α_i ?

The dual problem is

$$\max_{\alpha} \quad \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j$$

subject to

$$\sum_{i=1}^N \alpha_i y_i = 0 \quad \text{and} \quad 0 \leq \alpha_i \leq C \quad \forall i.$$

0
1

b) Given an optimal solution α^* , how do you recover $\mathbf{w}^*$? For which support vectors can you compute b^* , and why? **Justify your answer!**

The weight vector is recovered as

$$\mathbf{w}^* = \sum_{i=1}^N \alpha_i^* y_i \mathbf{x}_i.$$

To compute b^* , we can use any support vector with $0 < \alpha_i^* < C$. By KKT conditions, these points satisfy $\xi_i = 0$ and lie on the margin, hence

$$y_i(\mathbf{w}^{*\top} \mathbf{x}_i + b^*) = 1 \quad \Rightarrow \quad b^* = y_i - \mathbf{w}^{*\top} \mathbf{x}_i.$$

For points with $\alpha_i^* = C$, we can have $\xi_i > 0$ (margin violations), so they do not necessarily satisfy the margin equality and are not suitable to determine b^* .

0
1

0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

c) Consider the polynomial kernel $k(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^\top \mathbf{x}_2 + 1)^2$ for $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^2$. **Derive** an explicit feature mapping $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^M$ such that $k(\mathbf{x}_1, \mathbf{x}_2) = \phi(\mathbf{x}_1)^\top \phi(\mathbf{x}_2)$.

Expanding,

$$\begin{aligned} (\mathbf{x}_1^\top \mathbf{x}_2 + 1)^2 &= (x_{11}x_{21} + x_{12}x_{22} + 1)^2 \\ &= x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2x_{11}x_{12}x_{21}x_{22} + 2x_{11}x_{21} + 2x_{12}x_{22} + 1. \end{aligned}$$

We can write this as an inner product by choosing

$$\phi(\mathbf{x}) = \begin{pmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 \\ \sqrt{2} x_2 \\ 1 \end{pmatrix} \in \mathbb{R}^6.$$

Sample Solution

Problem 8 Dimensionality Reduction (4 credits)

0 ☐ Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be a data matrix of n samples with d features and $\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$ its empirical covariance matrix. We define the *total variance* of $\mathbf{X}$ as

$$\text{TVar}(\mathbf{X}) = \text{Tr}(\mathbf{\Sigma}) = \sum_{i=1}^n \Sigma_{ii}.$$

1 ☐ Let $\mathbf{\Gamma} \in \mathbb{R}^{d \times d}$ be an alternative basis of $\mathbb{R}^d$, i.e. its columns are pairwise orthogonal and of unit norm. **Prove** that the total variance of $\mathbf{X}$ does not change if we project our data onto the basis $\mathbf{\Gamma}$.

2 ☐ *Hint:* You can cycle matrices inside the trace operator, i.e. $\text{Tr}(\mathbf{ABC}) = \text{Tr}(\mathbf{BCA}) = \text{Tr}(\mathbf{CAB})$.

3 ☐ Let $\mathbf{Y} = \mathbf{X}\mathbf{\Gamma}$ be the projection of $\mathbf{X}$ onto the basis $\mathbf{\Gamma}$. The empirical covariance matrix of $\mathbf{Y}$ is

$$\mathbf{\Sigma}_Y = \frac{1}{n} \mathbf{Y}^\top \mathbf{Y} = \frac{1}{n} (\mathbf{X}\mathbf{\Gamma})^\top (\mathbf{X}\mathbf{\Gamma}) = \frac{1}{n} \mathbf{\Gamma}^\top \mathbf{X}^\top \mathbf{X} \mathbf{\Gamma} = \mathbf{\Gamma}^\top \mathbf{\Sigma} \mathbf{\Gamma}.$$

4 ☐ The total variance of $\mathbf{Y}$ is given by

$$\text{TVar}(\mathbf{Y}) = \text{Tr}(\mathbf{\Sigma}_Y) = \text{Tr}(\mathbf{\Gamma}^\top \mathbf{\Sigma} \mathbf{\Gamma}).$$

Using the cyclic property of the trace, we obtain

$$\text{Tr}(\mathbf{\Gamma}^\top \mathbf{\Sigma} \mathbf{\Gamma}) = \text{Tr}(\mathbf{\Gamma} \mathbf{\Gamma}^\top \mathbf{\Sigma}) = \text{Tr}(\mathbf{I} \mathbf{\Sigma}) = \text{Tr}(\mathbf{\Sigma}) = \text{TVar}(\mathbf{X}),$$

where we used that $\mathbf{\Gamma} \mathbf{\Gamma}^\top = \mathbf{I}$ since $\mathbf{\Gamma}$ is orthogonal.

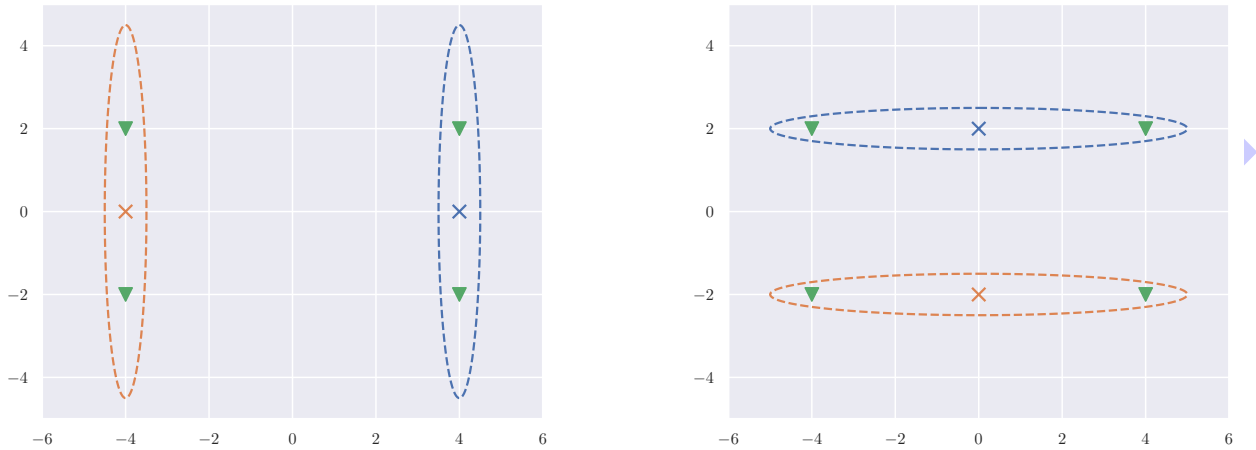
Problem 9 GMMs & EM (3 credits)

This problem is about Gaussian mixture models (GMM) and fitting them to data with the expectation maximization (EM) algorithm.

In this problem, the observed data are depicted as downwards pointing triangles.

a) In the figures below, sketch two possible optima that we can achieve when fitting a 2-component GMM to this data with EM. When you sketch components of a GMM, mark the component center with a cross (x) and indicate the shape of the component covariance matrix with an ellipse.

0
1



b) Consider the three 1-dimensional data points $\mathcal{D} = \{0, 1, 3\}$. We want to run K-means++ to initialize $K = 2$ centroids. What is the probability that the two centroids are initialized in different “clusters”, i.e. one centroid in the subset $\{0, 1\} \subset \mathcal{D}$ and one centroid in $\{3\} \subset \mathcal{D}$? **Justify your answer!**

0
1
2

With probability $\frac{1}{3}$, we select 3 as the initial centroid, so the second has to go in the other cluster.

With probability $\frac{1}{3}$, we select 0 as the initial centroid, and we select 3 with probability

$$\frac{3^2}{3^2 + 1^2} = \frac{9}{10}. \quad (9.1)$$

With probability $\frac{1}{3}$, we select 1 as the initial centroid, and we select 3 with probability

$$\frac{2^2}{2^2 + 1^2} = \frac{4}{5}. \quad (9.2)$$

This gives a total probability of

$$\frac{1}{3} \cdot \left(1 + \frac{9}{10} + \frac{4}{5}\right) = \frac{1}{3} \cdot \frac{10 + 9 + 8}{10} = \frac{27}{30} = \frac{9}{10}. \quad (9.3)$$

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large grid of graph paper, consisting of 30 columns and 30 rows of small squares. A diagonal watermark with the text "Sample Solution" in a light blue, sans-serif font is overlaid across the grid, running from the bottom-left towards the top-right.